

presimulate.pop.draws_cluster_documentation

Fred Cudhea

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R Markdown

Documentation for R script: `presimulate.pop.draws_cluster.r`, which creates simulation draws for population counts of the subgroups of interest based on estimates for mean and corresponding standard errors. I call it “pre”-simulate because I’m creating the draws before doing the simulating of PIFs and `impacts`. It’s useful to get these sims at the beginning (as opposed to redoing them when we calculate impacts for each outcome) so we can use a consistent set of sims we use for all impacts and outcomes.

Create variables to match the specific naming and coding of other input files. Note that “pop” is the dataframe with data on population counts. That data has already been read in in “`LASTING_cluster_w_master_input.r`”. Also, don’t be confused by the “pop” being essentially renamed to “mort”. This is actually just legacy from previous versions of the code which created mortality draws instead of population draws. I consider this code to be very “under the hood” so hopefully you don’t have to navigate this part of the code very often. That being said, if you make changes to the age/sex/race groups (or add/subtract strata), then you will need to change this part of the code to match.

```

##pre-simulate mort/incidence draws

#Library(Kmisc)

#setwd("C:\\Users\\jliu28\\Box Sync\\Juju and Fred\\Fangfang's cancer burden PAF")
#setwd("/Users/fcudhe01/Box Sync/Juju and Fred/Fangfang's cancer burden PAF")
#setwd(file_location)
#setwd("Inputs")
#setwd("C:\\Users\\fcudhe01\\Box Sync\\new Life\\Cancer-CRA-organized\\inputs")

#vars.to.keep<-c("disease", "Sex", "age_gp", "race_gp", "count", "se")
#vars.to.keep<-c("disease", "subgroup", "Sex", "Age", "Race", "count", "se")

#mort<-pop[,vars.to.keep]
mort<-pop

mort$agecat<-as.numeric(as.factor(mort$Age))

#####Edits made by Heesun#####
mort$female<-as.numeric(mort$Sex)
mort$female[as.numeric(mort$Sex)==2]<-0
#mort$female<-swap(as.numeric(mort$Sex)-1, from=c(0,1), to=c(1,0))

mort$race<-mort$Race
#mort$race <- as.numeric(as.factor(mort$Race))
### Other - 2, Black - 1, White - 3 ###
#mort$race[as.numeric(as.factor(mort$race_gp))==1]<-2
#mort$race[as.numeric(as.factor(mort$race_gp))==2]<-3
#mort$race[as.numeric(as.factor(mort$race_gp))==3]<-1
### Other - 3, Black - 2, White -1 ###
#mort$race<-swap(as.numeric(mort$race_gp), from=c(1,2,3,4), to=c(3,2,1,4))
#####

##add vars used for price/environment stuff##
mort <- mort %>% mutate(subgroup_id = subgroup,
                        age_gp = case_when(agecat==1 ~ '20-34',
                                           agecat==2 ~ '35-44',
                                           agecat==3 ~ '45-54',
                                           agecat==4 ~ '55-64',
                                           agecat==5 ~ '65-74',
                                           agecat==6 ~ '75+'),
                        #race_gp = case_when(Race_Label==1 ~ 'NHW',
                        #                      Race_Label==2 ~ 'NHB',
                        #                      Race_Label==3 ~ 'HIS',
                        #                      Race_Label==4 ~ 'OTH'),
                        sex_gp = case_when(female==0 ~ 'Male',
                                           female==1 ~ 'Female')
                        )

mort$race_gp <- mort$Race_label

mort$age<-0
mort$age[mort$agecat==1]<-25

```

```

mort$age[mort$agecat==2]<-35
mort$age[mort$agecat==3]<-45
mort$age[mort$agecat==4]<-55
mort$age[mort$agecat==5]<-65
mort$age[mort$agecat==6]<-75

mort$race<-0
mort$race[mort$Race_label=="NHW"]<-1
mort$race[mort$Race_label=="NHB"]<-2
mort$race[mort$Race_label=="HIS"]<-3
mort$race[mort$Race_label=="OTH"]<-4

```

Loop through each row and generate **population sims** based on mean and standard error of population for each subgroup.

```

observed.pop.draws<-matrix(data=NA, nrow=dim(mort)[1], ncol=nsim1)
for(i in 1:dim(mort)[1])
{
  temp<-rnorm(n=nsim1, mean=mort$population[i], sd=mort$population.se[i])
  temp[temp<0]<-0
  observed.pop.draws[i,]<-temp
}
observed.pop.draws<-cbind(mort, observed.pop.draws)

```

Save sims as a csv file.

```

#setwd("C:\\Users\\fcudhe01\\Box Sync\\new Life\\Cancer-CRA-organized\\inputs")
#write.csv(x=observed.pop.draws, file="in/TEMP/observed.pop.draws.csv")
#write.csv(x=observed.pop.draws, file="/cluster/tufts/lasting/shared/model_development/data/inputs/CRA_inpu
#write.csv(x=observed.pop.draws, file="/cluster/tufts/lasting/shared/model_development/data/inputs/observed
write.csv(x=observed.pop.draws, file=paste0(file_location, "/observed.pop.draws.csv"), row.names=FALSE)

```